

Department of Electrical and Electronics Engineering

University of Dhaka
Batch - 103, EEE-58

2nd Year 1st Semester



Course: EEE-2101

Title: Analog Electronics I

Class Teacher: Md. Shahin Parvej

Note by
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Session: 2023-2024
Class Roll: SH-042

SYLLABUS

PN Junction Diode and Diode Circuits: Semiconductor Materials, Construction of PN Junction, Formation of Depletion Layer and Barrier Voltage in PN Junction, Current-Voltage Characteristics of a PN Junction Diode, Equivalent Circuit of Diode, Resistance and Capacitance of PN Junction, Diode Circuit and Load Line, Zener Diode Operation and Application, Half-Wave and Full-Wave Rectifier, Voltage Multiplier, Clipper and Clamper.

Bipolar Junction Transistors (BJT): Construction and Operation of BJT, Amplifying Action, Characteristics of BJT in Common Base (CB), Common Emitter (CE) and Common Collector (CC) Configurations, α and β , Q-Point and Load Line, Different Biasing Circuits, Stability Factor, BJT as a Switch, re and Hybrid Equivalent Circuit of BJT.

Single Stage BJT Amplifier Circuits: Operation of Single-Stage Amplifier, Voltage and Current Gain, Input and Output Impedance of CB, CE, CC Configurations using h-Parameter, Effect of Unbypassed R_E , R_S .

Field Effect Transistor (FET): JFET Structure, Operation and Characteristics, hParameters for JFET, MOSFET Construction, Operation and Characteristics, Biasing Circuits of JFET and MOSFET, Single-stage JFET Amplifier, MOSFET as a Switch, CMOS Inverter.

Multistage and Differential Amplifiers: RC Coupled Two Stage Amplifiers, Direct-coupled Amplifier, Darlington Pair, Multistage Frequency Effects.

Feedback Amplifiers: Principle of Feedback Amplifier, Positive and Negative Feedback, Advantages of Negative Feedback, Gain Stability, Decreased Distortion, Increased Bandwidth, Forms of Negative Feedback, Practical Feedback Circuits.

REFERENCE BOOKS

1. Electronic Devices and Circuits, 5th Edition, David Bell, Oxford University Press.
2. Electronic Devices and Circuit Theory, Paperback Edition, R. Boylestad and L.Nashelsky, Pearson.
3. Electronic Devices and Circuits, Allen Mottershead, Goodyer Pub. Co.
4. Electronic Principles, Paperback Edition, Albert Malvino and David Bates, McGraw Hill.
5. Electronic Circuits: Discrete and Integrated, Paperback Edition, Donald Schilling, Charles Belove, and Raymond Saccardi, McGraw Hill.
6. Microelectronics, 2nd Edition, Jacob Millman and Arvin Grabel, McGraw Hill.
7. Microelectronic Circuits: Theory and Applications, 7th Edition, Adel S. Sedra, Kenneth C. Smith and Arun N. Chandorkar, Oxford University Press.

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CHAPTER 1

PN JUNCTION DIODE AND DIODE CIRCUITS